13.11 What Is the Figure?

Lesson Introduction

Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	Learning Targets	• I can use coordinate geometry to classify triangles and quadrilaterals. • I can use coordinate geometry to justify definitions, properties, or theorems of triangles and quadrilaterals. • I can use coordinate geometry to solve mathematical problems involving triangles and quadrilaterals.

Benchmark Coherence	Building On	Working Toward
	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.	N/A

Essential Questions	• How can we use a quadrilateral's property to help justify that we have correctly identified them? • What can help us distinguish the difference between multiple quadrilaterals?
Lesson Narrative	In this lesson, students continue to use coordinate geometry to justify and classify polygons based on their properties through a Card Sort. Students are tasked to match each image with its correct classification, coordinates, and property, while also justifying their matches.

Component Summary				
	13.11.1 Warm-Up (~10 mins)	Bell Work *(SE p. 345)*	13.11.2 Activity (35 mins)	Card Sort *(SE p. 346)*
	13.11.3 Wrap-Up (~5 mins)	Students summarize their learning *(SE p. 347)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Glossary: N/A	• 13.11.2 Activity Card Sort cards • Calculators	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may get the properties mixed up because some properties of polygons overlap.
Students may miscalculate the length measurements when using the Pythagorean Theorem or the Distance Formula.
Students may complete matches by the look of the polygon, instead of justifying with math.</td><td>Consider having students who are more visual learners to have graph paper or a graphing tool for the 13.11.1 Warm-Up.
Consider having an anchor chart with the various type of quadrilaterals and polygons, along with an image for students who need help, during the Card Sort when matching images with names of polygons.
Consider using information from the previous lessons to form a small group of students who are struggling with using coordinate geometry to classify figures. Facilitate these students as they work through 13.11.2 Activity.
If time permits, consider utilizing 13.11.2 Activity over two days. The content that is spiraled in the activity utilizes concepts found in the three benchmarks noted, as well as concepts from MA.912.GR.1.4 and MA.912.GR.1.5.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Anticipate, Monitor, Select, Sequence, Connect
In the 13.11.12 Card Sort, students are anticipating, sequencing, and making connections between the multiple representations of polygons and how they are each represented. Students create a sequence on how they organize and choose their cards, and can form the connection between these representations through their justifications.</td><td>MA.K12.MTR.2.1: Multiple Representations
In the 13.11.2 Card Sort, students are building their understanding of coordinate geometry through using graphs, coordinate points, and properties of polygons. Students are expressing these connections between the concepts and representations using justifications.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In the 13.11.1 Warm-Up, consider that visual learners may have difficulties with identifying the stage that is in the shape of a trapezoid, so consider providing a graphing tool or graph paper for those students. Throughout the entire lesson, also consider having an anchor chart near the front of the class where students can reference the various types of quadrilaterals and polygons.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.11.1 Warm-Up: Bell Work

Component Overview: *Students are comparing two different quadrilateral stages and determining which one to use as a stage.*

13.11.1 Warm-Up: Bell Work

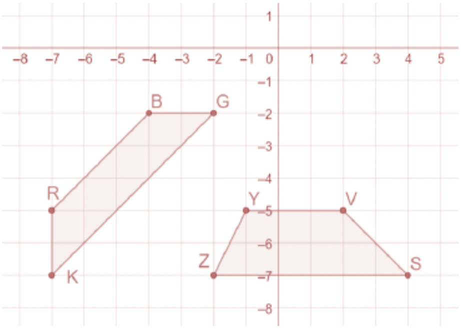
Jonah is designing a stage in the shape of a trapezoid. Jonah has created two designs.

$VSZY$	$BGKR$
$V(2,-5)$, $S(4,-7)$, $Z(-2,-7)$, and $Y(-1,-5)$	$B(-4,-2)$, $G(-2,-2)$, $K(-7,-7)$, and $R(-7,-5)$

1. Determine whether Jonah can use either or both designs for the trapezoidal stage.

    - Show your work in an organized way.
    - Write your conclusion explaining the properties used to determine your answer.

    Answers will vary. Both shapes create a trapezoid. The definition states that a quadrilateral is a trapezoid if at least one pair of sides are parallel. In trapezoid $RBGK$ both $\overline{RB}$ and $\overline{KG}$ have slopes equal to 1, making them parallel. In trapezoid $YVSZ$ both $\overline{YV}$ and $\overline{ZS}$ have slopes equal to 0, making them parallel.



Use student understanding to inform your instruction, and to determine where further student support could be implemented. Make a list of misconceptions, and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.

Consider that visual learners may have difficulties with identifying the stage that is in the shape of a trapezoid, so consider providing a graphing tool or graph paper for those students.

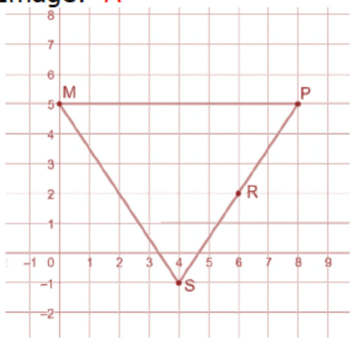
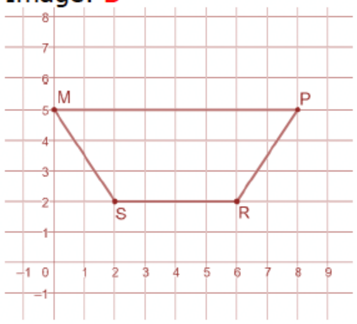
Consider also having an anchor chart near the front of the class where students can reference the various types of quadrilaterals, especially when trying to determine which stage Jonah should pick.

13.11.2 Activity: Card Sort

Component Overview: *Students are completing a Card Sort where they are matching each image, classification, and property that represents the same polygon. To make sure students are sure in their choices, they are then asked to provide justification with each match.*

13.11.2 Activity: Card Sort

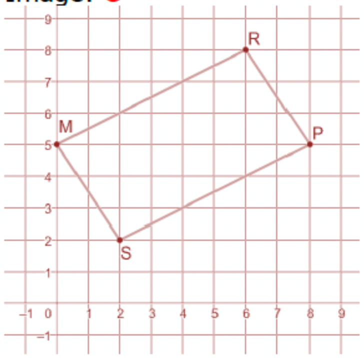
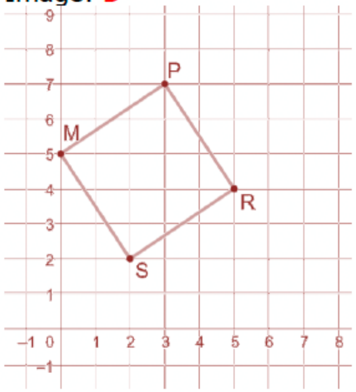
Match each image, classification, coordinates, and property that represent the same polygon. Record your matches in the table by writing the letter or number of each card and showing your justifications.
Order of matches may vary. Both the letter/number and answer are shown below.

Match 1	Match 2
Image: A 	Image: B 
Coordinates: 16 M(0,5), P(8,5), R(6,2), and S(4, −1)	Coordinates: 13 M(0,5), P(8,5), R(6,2), and S(2,2)
Classification: 4 *Isosceles triangle*	Classification: 7 *Isosceles trapezoid*
Property: 21 *Two adjacent sides are congruent*	Property: 23, 26 or 30 and 34 (also 22) *Diagonals are congruent. One pair of sides are congruent.*
Justification: *Answers will vary.* SP and $MS = \sqrt{52}$	Justification: *Answers will vary.* RP and $SM = \sqrt{13}$ SP and $RM = \sqrt{45}$

Students will match each image, classification, coordinate, and property(s) that represents the same polygon. Students record their matches in the table by writing the letter or number in their workbook. Consider having students work with a partner or small group. Ten cards are provided. This provides you with flexibility to determine how best to use the cards for your students. If students are struggling with certain figures, consider choosing to focus on these figures. A few different ways to utilize the Card Sort are:

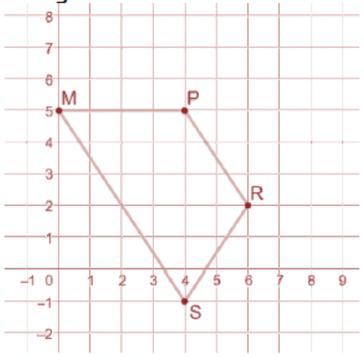
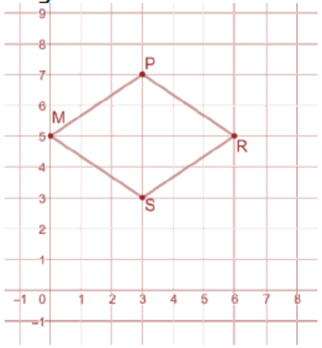
- *Have students work individually on 2-3 cards that represent figures your students have struggled with. Then, have students check their work with a partner. Allow students to adjust their answers. Then, consider giving students a copy of the remaining cards to complete on their own.*
- *Have students work in small groups on 4-5 cards that your students struggled with the most. Then, consider providing students a copy of the remaining cards to complete on their own.*
- *Utilize stations where the image or the set of coordinates is the station's anchor, and each station has all of the classifications, properties, coordinates or images, and justifications to sort through.*
- *Utilize the cards as a Gallery Walk where the image or the set of coordinates is displayed. Students then sort through the cards to add one to what is already displayed. Once 4 rotations are completed, then students rotate to verify the posted cards from the previous groups, adding a post-it note that states their agreement or their suggested change.*

13.11.2 Activity: Card Sort (continued)

Match 3	Match 4
Image: C 	Image: D 
Coordinates: 19 M(0,5), P(8,5), R(6,8), and S(2,2)	Coordinates: 17 M(0,5), P(3,7), R(5,4), and S(2,2)
Classification: 10 *Parallelogram*	Classification: 9 *Square*
Property: 25, 35 *Diagonals bisect each other.* *Opposite sides are congruent.* **Card 35 is also sufficient on its own.*	Property: 23, 26 or 30 and 27, 29, or 32 *Diagonals are congruent.* *Diagonals are perpendicular.*
Justification: *Answers will vary.* RM *and* $PS = \sqrt{45}$ MS *and* $RP = \sqrt{13}$ *Midpoint of* $\overline{MP}$ *is* $(4,5)$ *Midpoint of* $\overline{RS}$ *is* $(4,5)$	Justification: *Answers will vary.* MR *and* $PS = \sqrt{26}$ *Slope of* $\overline{MR} = -\frac{1}{5}$ *Slope of* $\overline{PS} = 5$

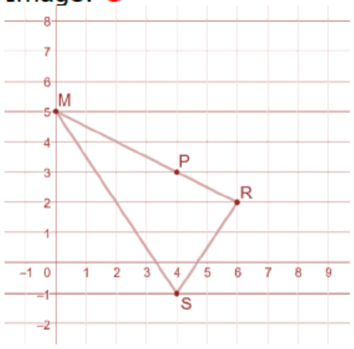
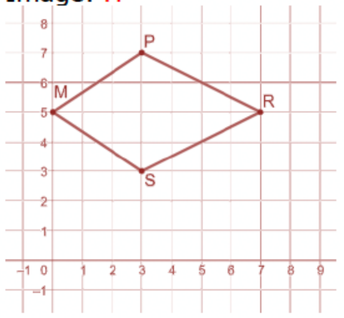
Consider having an anchor chart near the front of the class with the name and image of a variety of polygons and quadrilaterals. This can be especially helpful to visual learners. To challenge students, consider removing the image from the Card Sort.

13.11.2 Activity: Card Sort (continued)

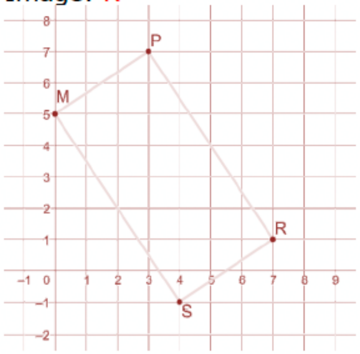
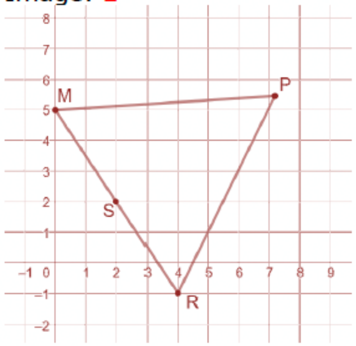
Match 5	Match 6
Image: E 	Image: F 
Coordinates: 11 *$M(0,5)$, $P(4,5)$, $R(6,2)$, and $S(4,-1)$*	Coordinates: 20 *$M(0,5)$, $P(3,7)$, $R(6,5)$, and $S(3,3)$*
Classification: 3 *Trapezoid*	Classification: 5 *Rhombus*
Property: 22 *One pair of parallel sides.*	Property: 27, 29, or 32 and 28 or 31 *Diagonals are perpendicular. All sides are congruent.*
Justification: *Answers will vary.* *Slope of $\overline{PR} = -\frac{3}{2}$* *Slope of $\overline{MS} = -\frac{3}{2}$*	Justification: *Answers will vary.* *Slope of $\overline{MR} = 0$* *Slope of $\overline{PS}$ is undefined* $MP = PR = RS = SM = \sqrt{13}$

Consider that students will get match 4 (square) and match 6 (rhombus) mixed up, since, visually, they can look similar. Consider making sure students calculate the slope and length of the diagonals in their justification, so that they are sure about their matches.

13.11.2 Activity: Card Sort (continued)

Match 7	Match 8
Image: G 	Image: H 
Coordinates: 14 *$M(0,5)$, $P(4,3)$, $R(6,2)$, and $S(4,-1)$*	Coordinates: 18 *$M(0,5)$, $P(3,7)$, $R(7,5)$, and $S(3,3)$*
Classification: 1 *Scalene triangle*	Classification: 8 *Kite*
Property: 24 *No sides are equal.*	Property: 27, 29, or 32 and 33 *Diagonals are perpendicular. Exactly one diagonal bisects the other.*
Justification: *Answers will vary.* $MR = \sqrt{45}$ $RS = \sqrt{13}$ $MS = \sqrt{52}$	Justification: *Answers will vary.* *Slope of $\overline{PS}$ is undefined* *Slope of $\overline{MR} = 0$* *Midpoint of $\overline{PS}$ is $(3,5)$* *Midpoint of $\overline{MR}$ is $(3.5,5)$* *The equation of line $\overline{MR}$ is $y = 5$ and the midpoint of $\overline{PS}$ is a point on that line.* *The equation of line $\overline{PS}$ is $x = 3$, but the midpoint of $\overline{MR}$ does not lie on that line.*

- *How are match 6 (rhombus) and match 8 (kite) different?*
- *How can we determine the difference between the two polygons?*

Match 9	Match 10
Image: K 	Image: L 
Coordinates: 15 $M(0,5)$, $P(3,7)$, $R(7,1)$, and $S(4,-1)$	Coordinates: 12 $M(0,5)$, $P(7.5,5.5)$, $R(3.3,11.7)$, and $S(2,9)$
Classification: 2 *Rectangle*	Classification: 6 *Equilateral triangle*
Property: 23, 26 or 30 *Diagonals are congruent*	Property: 28 or 31 *All sides are congruent*
Justification: *Answers will vary.* $RM = PS = \sqrt{65}$	Justification: *Answers will vary.* $RM = PR = PM = \sqrt{56.5}$

Consider that students will get match 10 (equilateral triangle) and match 1 (isosceles triangle) mixed up, since, visually, they can look similar. Consider making sure students calculate the side lengths in their justification, so that they are sure about their matches.

Consider that students will get match 3 (parallelogram) and match 9 (rectangle) mixed up, since, visually, they can look similar. Consider making sure students look at the diagonals in their justification, so that they are sure about their matches.

- *How are match 3 (parallelogram) and match 9 (Rectangle) similar?*
- *How are they different?*
- *How can we tell the difference between them?*

For students who finish early, challenge them to find 3 different matches that they think other students in the class may have difficulties with and write an explanation why.

13.11.3 Wrap-Up: Cool Down

Component Overview: *Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

13.11.3 Wrap-Up: Cool Down

1. Explain how you can use coordinate geometry to classify triangles and quadrilaterals.

   Answers will vary.